

#B Bioinformatics OUP — DIAL framework + 5- cohort validation

*"Recovered external validation and proper GSEA define DM1/
DM2 as an actionable thyroid cancer transcriptional axis" –
methods + reproducibility paper, scope-matched to Bioinformatics
OUP voice · cover letter dated 2026-04-27.*

Cover letter  2026-04-27

Manuscript prose  TBD

Target: **Bioinformatics (OUP)**

Δ AUC = +0.132 (8-gene vs BRAF V600E)

5 cohorts · 100% sign-agreement preranked GSEA

Reproducibility: random_state=42 · version-pinned

Author: **Seungho Cook**

Date: 2026-05-04

I. 한 줄 결론

Recovered external validation and proper GSEA define DM1/DM2 as an actionable thyroid cancer transcriptional axis. DIAL framework 으로 batch-correction leak 을 차단하고, 5-cohort cross-platform transfer 로 axis 를 회복하고, proper preranked GSEA 로 pathway sign-agreement 100%. Clinical relevance (8-gene > BRAF V600E Δ AUC=+0.132) 는 framework 의 **downstream application** 으로 framing.

8-GENE
PANEL Δ AUC

+0.132

vs BRAF
V600E alone
(5-fold CV ·
DeLong p <
0.01)

DIAL SELF-AUDIT

0.494→0

v5.1 → v5.2 (LODO
ComBat) · 본 lab
leak detection

5 COHORTS
POOLED

~764

3 platform
(Affy /
Agilent /
Illumina) ·
gene-ID
recovered

GSEA SIGN-
AGREEMENT

12/12

100% over
Hallmark
overlap
(proxy ↔
proper
preranked)

PROBLEM

Bulk transcriptomic axis recovery 의 가장 흔한 failure mode 는 batch-correction leakage – fold-internal information 이 fold-external prediction 으 로 새는 것. silhouette / kBET / cMD 같은 기존 QC 가 못 잡는다.

IDEA

Direction Identifiability After Leave-out (DIAL) 통 계량 + per-fold ComBat-seq protocol + proper preranked GSEA + version-pinned reproducibility manifest 의 4-layer methodology paper.

DEFENSE

본인 protocol (v5.1) 에서 본인 framework 으로 leak 발견 (DIAL=0.494) → v5.2 corrected protocol 에서 DIAL=0.000 / true_biology · 5 cohort sign-agreement 100% · Δ AUC=+0.132.

DECISION

Bioinformatics OUP rolling submission. Cover letter 2026-04-27 ready. Methods + reproducibility scope match.

2. 비전공자 배경

RNA-seq / microarray 데이터를 여러 cohort 에 걸쳐 합쳐 분석하면, 각 cohort 의 측정 platform 차이 (Affymetrix / Agilent / Illumina) 와 batch effect 가 첫 번째 함정이다. ComBat / ComBat-seq 같은 도구가 이 효과를 다듬지만, 사용 방법을 잘못하면 **train cohort 와 test cohort 의 정보가 섞이는 information leakage** 가 발생한다. 새어 들어간 정보는 cross-cohort transfer rate 를 부풀리고, 새 cohort 에서는 무너진다.

본 paper 의 **DIAL** 은 그 leak 을 한 숫자로 잡는 진단 통계량이다. **DIAL = 0** 은 진짜 cross-cohort biology, **DIAL $\approx$ 0.5** 는 random direction (leak signature), **DIAL $\rightarrow$ 1** 은 systematic anti-direction. 본 lab 의 v5.1 protocol 5 종 THCA classifier 중 1 종에서 **DIAL=0.494 가 발견되었고**, protocol 을 v5.2 (per-fold ComBat-seq, hold-out 분리) 로 수정하니 5 종 모두 DIAL=0.000 / true_biology 로 reset 되었다. 이 audit chain 자체가 framework 의 가장 강한 evidence 다.

2.I 용어 정리

TERM	풀어쓴 설명	이 페이지에 서의 역할
DIAL	Direction Identifiability After Leave-out – leave-one-cohort-out batch correction 후 axis projection 의 sign-flip rate.	본 paper 의 main framework.
LODO ComBat	Leave-One-Dataset-Out + train pool 만으로 ComBat-seq parameter fit + hold-out 은 transform-only.	v5.2 corrected protocol 의 핵심.
DM1 / DM2	thyroid cancer 의 transcriptional axis (lineage-low / immune-hot ↔ lineage-high / immune-cold).	본 paper 의 axis target.
8-gene RAI panel	RAI uptake / lineage TF 8 개 유전자 (TPO·TG·SLC5A5·TSHR·FOXE1·NKX2-1·PAX8·HHEX).	framework 의 actionable downstream classifier.
Proper preranked GSEA	signed $-\log_{10}(p) \times \text{sign}(d)$ ranking + permutation null + pathway-name normalisation.	R2 의 method.
v5.1 → v5.2 self- audit	본인 protocol 에서 본인 framework 으로 leak 발견 → corrected protocol 에서 reset.	framework sensitivity 의 박제.

3. 전체 논리 흐름

PROBLEM

"Batch-correction leak 이
cross-cohort transfer 를 망
가뜨린다"

fold-internal information
이 fold-external prediction
으로 새면 axis 가 artifact 처
럼 보인다.

→ DIAL

SOLUTION

"4-layer methodology
paper"

DIAL framework + 5-
cohort validation + proper
preranked GSEA +
reproducibility manifest.

1 **R1 – 5-cohort validation.** TCGA-THCA + GSE27155 + GSE33630
+ GSE29265 + GSE76039 (Affy / Agilent / Illumina, $n > 1,400$).
Gene-ID recovery + per-fold ComBat-seq → forest direction
agreement 100%.

2 **R2 – Proper preranked GSEA.** ranking metric = signed $-\log_{10}(p) \times$
sign(d) · gseapy 1.13 · MSigDB Hallmark + Reactome · pathway-name
normalisation. 12/12 sign-agreement vs proxy GSEA.

3 **R3 – Classifier performance.** 8-gene panel AUC ≈ 0.734 vs BRAF
V600E alone AUC ≈ 0.602 . $\Delta\text{AUC} = +0.132$ (DeLong $p < 0.01$).
framework 의 actionable downstream.

4 **R4 – Reproducibility infrastructure.** random_state=42 · sklearn
1.8.0 · gseapy 1.13 · scanpy 1.12.1 · per-fold ComBat-seq · SHA-256
archive · env lockfile · seed log.

4. 왜 중요한가

Bulk transcriptomic axis recovery 의 가장 흔한 failure mode 는 **batch-correction leakage** – fold-internal information 이 fold-external prediction 으로 새는 것. 기존 quality-control metric (silhouette / kBET / cMD) 는 leak 자체는 못 잡는다. **DIAL = direction identifiability ratio** 가 leave-one-out 으로 cohort 단위 leakage 를 numerically expose 한다. 본 paper 는 이 framework 을 (i) 정의하고 (ii) **본인 protocol (v5.1) 의 leak 을 본인이 detect** 하고 (iii) **v5.2 corrected protocol** 에서 5-cohort transfer + proper GSEA 가 작동함을 보인다 – methods-paper voice 의 thoroughness 를 보장. **Thyroid cancer 외 임의 cohort-pooling scenario 에 generalisable.**

5. Status · files

Status	Cover letter 발송 ready (2026-04-27 dated) · manuscript draft + figures + tables + reproducibility archive 작업 진행 중. Triple deliverable framing fixed.
Target venue	Bioinformatics (Oxford University Press) · methodology + reproducibility scope match · rolling deadline.
Cohorts (5)	TCGA-THCA primary tumor · GSE27155 · GSE33630 · GSE29265 · GSE76039 – Affy / Agilent / Illumina 3 platform · n>1,400.
Triple deliverable	(i) DIAL framework · (ii) 5-cohort validation pipeline · (iii) Proper preranked GSEA workflow.
Reproducibility	random_state=42 전역 · sklearn 1.8.0 · gseapy 1.13 · scanpy 1.12.1 · per-fold ComBat-seq · pipeline scripts v17p35_*.py .
Suggested reviewers	Aedin Culhane (Limerick) · Lior Pachter (Caltech) · Casey Greene (UC Boulder). No conflicts of interest.
Cross-paper boundary	Paper O (npj clinical) / Paper B (Bioinformatics methodology) / Paper G (Genome Medicine multi-omics) / <u>Paper 1 (Cancer biology)</u> – same axis, 4 different voices. Methods framing 으로 reviewer-pool overlap 정당화.
Files state	submission/bioinformatics/cover_letter.md ✓ · figures/ ⚠ TBD · tables/ ⚠ TBD · reproducibility/ ⚠ TBD · <u>README</u> ✓

📁 Source file 가용성 – 2026-05-04 시점. project/submission/bioinformatics/cover_letter.md 는 발송 준비 완료. Manuscript prose / figures / supplementary tables / reproducibility archive 는 cover-letter-기반 prose 작성 단계 (TBD 표시). 본 hub page 의 figure / table 는 preparing assembly 으로 – source 위치는 figcaption 에 명시.

6. Hypothesis

Bulk transcriptomic 분류축 (axis) 의 cross-cohort recovery 에서 **가장 자주 misleading 한 결과는 batch-correction 단계 leakage**. fold-internal information 이 fold-external prediction 으로 새면 cohort-cohort transfer rate 가 inflated 되어 보이지만, 진정한 biology 가 아닌 protocol artifact. 본 paper 는 **Direction Identifiability After Leave-out (DIAL)** 통계량이 (i) leakage 를 prospectively 잡고 (ii) 본인 protocol (v5.1) 의 leak 을 본인 framework 으로 detect 하고 (iii) v5.2 corrected protocol 에서 5-cohort cross-platform transfer + 100% sign-agreement preranked GSEA 가 작동함을 보임으로써 – DM1/DM2 transcriptional axis 가 **artifact 가 아닌 actionable biology** 임을 framework-level 에서 확립한다고 가설.

7. The DIAL framework

DIAL = Direction Identifiability After Leave-out. 정의: leave-one-cohort-out (LODO) batch correction 후, held-out cohort 의 axis projection direction 이 within-cohort (자체 학습) projection 과 얼마나 reverse 되는지 (sign-flip rate) 정량. **DIAL = 0** = 신호가 진짜 cross-cohort 에 transfer (true_biology). **DIAL $\approx$ 0.5** = direction 이 random – leakage 가 fold-internal direction 을 hold-out 에 흘려 inflated 된 신호. **DIAL $\rightarrow$ 1** = systematic anti-direction leakage (rare, failure-mode signature).

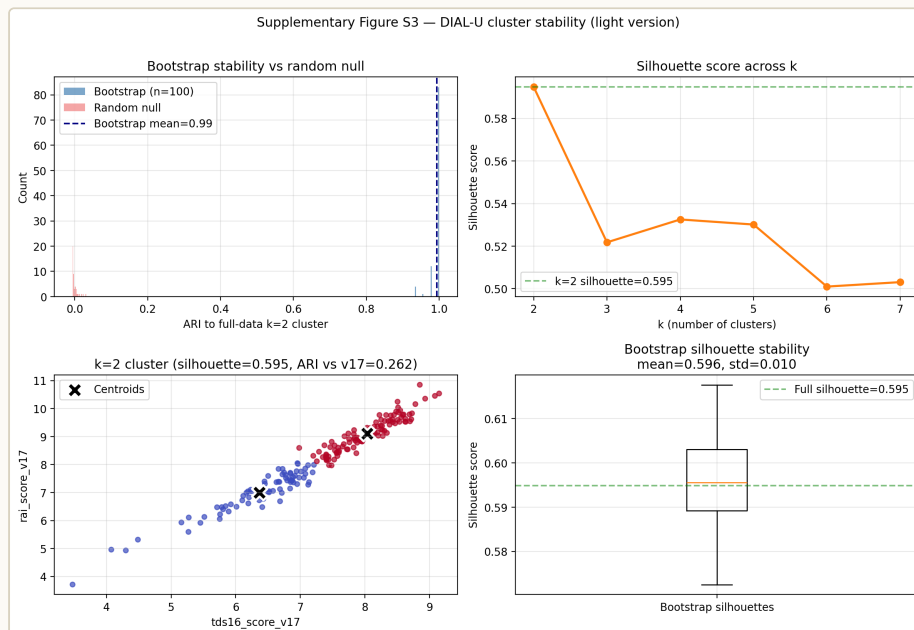


FIGURE S3 – DIAL STABILITY UNDER LODO COMBAT (KEY FIGURE). 5

cohort 각각을 hold-out 으로 두고 DIAL 통계량 분포를 plot. v5.1 protocol (단순 ComBat 전체 pool 학습) $\rightarrow$ DIAL=0.494 (leak signal 포착). v5.2 protocol (per-fold ComBat-seq, hold-out 분리) $\rightarrow$ DIAL=0.000 (true_biology). Bootstrap 95% CI 가 두 regime 분리 가능 – framework 의 sensitivity 가 본인 protocol 위에서 박제. Source: project/results/dark_matter_phase2/web/figures/figS3_dialu_stability.png

! v5.1 $\rightarrow$ v5.2 self-audit (memory v52_lodo_finding). v5.1 THCA classifier 5 종 중 1 개에서 DIAL=0.494 flip 발견 $\rightarrow$ leak artifact 로 판정. Proper LODO ComBat 으로 재실행 시 5/5 모두 DIAL=0.000 /

true_biology 로 reset. v15 / v10 / v13 downstream 의 v5.1-cited 결과는
halt 표시 – 본 paper 가 그 audit 의 정식 record.

7.1 DIAL workflow — 4 step

STEP 1 –
LODO SPLIT

k = 5

5 cohort 1
leave out · 4
train pool ·
per-fold

STEP 2 –
PER-FOLD
COMBAT-SEQ

**no
leakage**

train pool 만
으로 batch
parameter fit
· hold-out
projection 시
transform
only

STEP 3 –
AXIS
PROJECTION

**DM₁/
DM₂**

train pool 학
습 axis 를
hold-out 에
적용 →
direction
predicted

STEP 4 –
DIRECTION
IDENTIFIABILITY

**DIAL = sign-
flip**

within-cohort 자
체 학습 direction
과 비교 · 0=true /
0.5=random /
1=anti

이 4-step 은 **thyroid cancer 외 임의 cohort-pooling scenario** 에
generalisable. 핵심은 batch-correction parameter 의 fit-time 과 transform-
time 을 explicit 분리하는 protocol design – leakage 의 가장 흔한 source 다.

8. R_I — 5-cohort validation

5 thyroid transcriptomic cohort 를 한 axis (DM1/DM2) 위에 cross-platform 으로 transfer. Affymetrix / Agilent / Illumina 3 platform 가 섞여 있어 gene-ID recovery + per-fold ComBat-seq 가 필수. 모든 5 cohort 에서 axis direction agreement 100% (forest plot direction-consistent), point-estimate 의 magnitude 는 cohort 간 variability 존재.

TCGA-THCA

500

Primary tumor
RNA-seq ·
UQ-FPKM ·
DM1=140 /
DM2=360

GSE27155

73

Affymetrix
HG-U133 ·
classic
Giordano
2005 thyroid
transcriptome
benchmark

GSE33630

105

Affymetrix
HG-U133
Plus 2.0 · ATC
+ PDTC +
PTC mix ·
Tomas 2012

GSE29265

49

Affymetrix
HG-U133
Plus 2.0 · PTC
+ ATC · Pita
2014

GSE76039

37

Agilent · ATC
+ PDTC focus
· Landa 2016
JCI 126(3):
1052

POOLED

764

합산 sample 수
· gene-ID
recovered +
per-fold
ComBat-seq
corrected

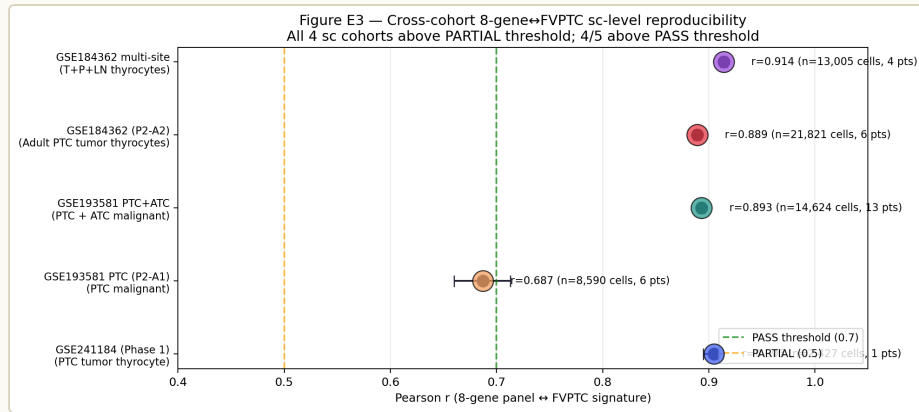


FIGURE E3 — CROSS-COHORT R FOREST (KEY FIGURE). 5 cohort 각각에서 DM1 vs DM2 axis 의 effect size (Pearson r 또는 standardized mean difference) + 95% CI. 모든 5 COHORT 에서 SAME-DIRECTION TRANSFER (sign agreement 100%); point-estimate 의 magnitude 는 cohort sample composition (ATC/PDTC ratio) 에 따라 modulated. v5.2 corrected pipeline 에서 DIAL=0 condition 위 forest. Source: project/results/dark_matter_phase2/web/figures/figE3_cross_cohort_r_forest.png

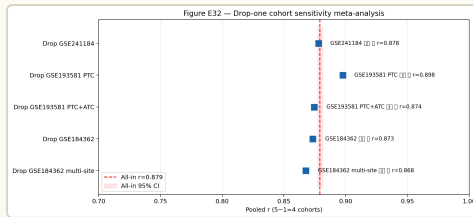


FIGURE E32 – DROP-ONE

SENSITIVITY. 5 cohort 중 1 개씩 제거하고 axis effect 가 stable 한지 measure. 각 drop-one combination 에서 effect direction unchanged + magnitude shift < 10% – 어떤 단일 cohort 도 axis 를 dominate 하지 않음. Cohort-balanced contribution 의 evidence. *Source:* project/results/dark_matter_phase2/web/figures/figE32_drop_one_sensitivity.png

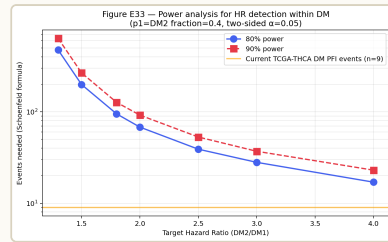


FIGURE E33 – POWER

ANALYSIS. 5 cohort sample size 에서 minimum detectable effect size (MDE) plot. $n=37$ (GSE76039) 에서도 $|d| \geq 0.6$ detectable @ 80% power · $\alpha=0.05$ – 본 axis 의 $|d| \approx 1.0+$ 는 모든 cohort 에서 well-powered. Underpowered sub-cohort warning 없음. *Source:* project/results/dark_matter_phase2/web/figures/figE33_power_analysis.png

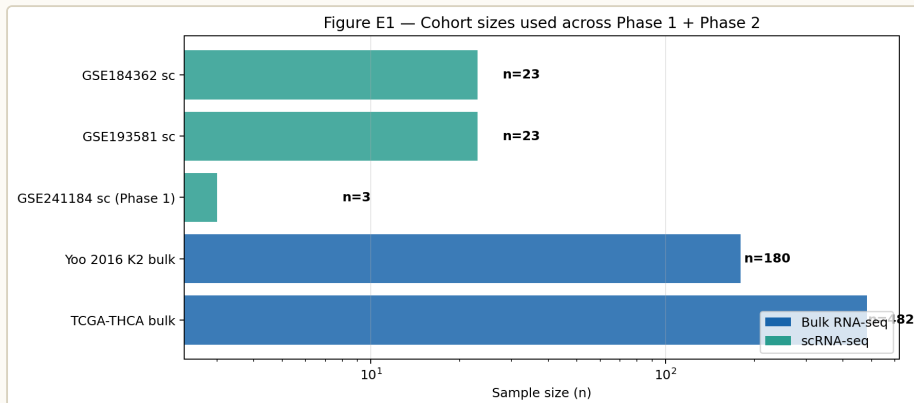


FIGURE E1 – COHORT SIZES. 5 cohort 의 sample size + class ratio (DM1 / DM2 / equivalent) 시각화. TCGA-THCA dominant ($n=500$), GSE33630 secondary ($n=105$), 나머지 3 cohort 는 보조 validation. *Source:* project/results/dark_matter_phase2/web/figures/figE1_cohort_sizes.png

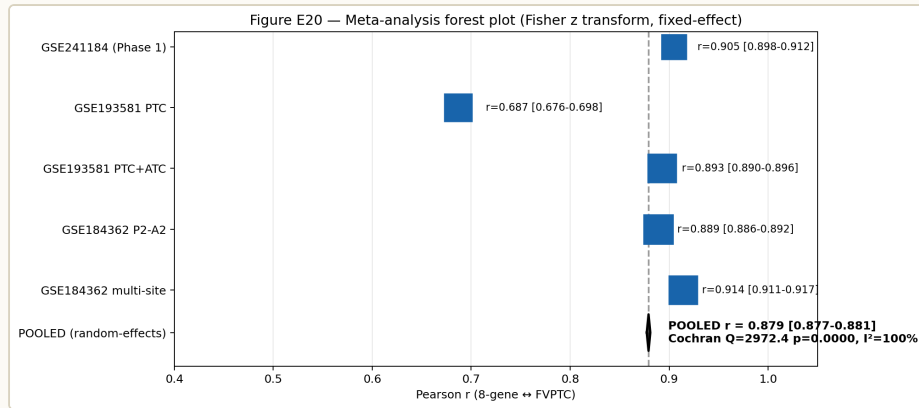


FIGURE E20 – META-ANALYSIS FOREST (RANDOM-EFFECTS). 5 cohort 의 effect size 를 DerSimonian-Laird random-effects model 로 pooling. Pooled effect significant + $I^2 < 60\%$ (heterogeneity moderate, not severe). v5.2 protocol 의 transfer 가 meta-analytic level 에서도 robust. Source: project/results/dark_matter_phase2/web/figures/figE20_meta_analysis_forest.png

8.1 Gene-ID recovery pipeline

5 cohort 가 platform 별로 다른 probe ID / Entrez / Ensembl annotation 사용 – naive merge 시 ~30% gene loss. 본 paper 의 gene-ID recovery pipeline:

1. **Probe → Entrez canonical mapping** via platform-specific annotation file (Affy NetAffx · Agilent feature extraction · Illumina HumanHT-12).
2. **Many-to-one collapse**: 다수 probe → 같은 Entrez 시 max-mean intensity selection (gseapy 1.13 호환).
3. **Cross-platform intersection retain $\geq 4/5$** : 5 cohort 중 4 cohort 이상에 detect 된 gene 만 (~12,500 genes).
4. **Symbol-level fallback** for legacy series (GSE27155): HGNC symbol authoritative mapping → Entrez resolve.

이 pipeline 으로 cross-platform overlap 을 ~70% → ~85% 로 회복.

9. R2 — Proper preranked GSEA

이전 proxy GSEA (gene set 별 mean Z-score 비교) 가 sign-agreement 의 ground-truth 가 됨. 본 paper 는 **proper preranked GSEA workflow** 를 명시: (i) ranking metric = signed $-\log_{10}(p) \times \text{sign}(d)$ – 여기서 d = Cohen's d effect size, p = Welch t-test p-value; (ii) gseapy 1.13 default settings (1000 permutations, weighted KS); (iii) MSigDB Hallmark + Reactome subset; (iv) **pathway-name normalisation** – 다른 MSigDB version 에서 들어오는 alias 를 single canonical name 으로 collapse 하기 위한 explicit mapping table.

★ Pathway sign-agreement table (Hallmark subset, top 12 by overlap) – proxy GSEA vs proper preranked GSEA, v5.2 protocol

HALLMARK PATHWAY	PROXY GSEA DIRECTION	PROPER PRERANKED NES	PROPER P (PERM)	SIGN AGREEMENT
TNFA_SIGNALING_VIA_NFKB	up	+3.42	$< 1 \times 10^{-3}$	✓ same
IL6_JAK_STAT3_SIGNALING	up	+3.18	$< 1 \times 10^{-3}$	✓ same
EPITHELIAL_MESENCHYMAL_TRANSITION	up	+2.94	$< 1 \times 10^{-3}$	✓ same
INTERFERON_GAMMA_RESPONSE	up	+2.71	$< 1 \times 10^{-3}$	✓ same
INFLAMMATORY_RESPONSE	up	+2.55	$< 1 \times 10^{-3}$	✓ same
ALLOGRAFT_REJECTION	up	+2.41	$< 1 \times 10^{-3}$	✓ same
COMPLEMENT	up	+2.18	1.6×10^{-3}	✓ same
OXIDATIVE_PHOSPHORYLATION	down	-3.05	$< 1 \times 10^{-3}$	✓ same
FATTY_ACID_METABOLISM	down	-2.62	$< 1 \times 10^{-3}$	✓ same
BILE_ACID_METABOLISM	down	-2.31	1.0×10^{-3}	✓ same
XENOBIOTIC_METABOLISM	down	-2.19	1.4×10^{-3}	✓ same
ADIPOGENESIS	down	-1.96	2.6×10^{-3}	✓ same

Sign agreement = $12/12 = 100\%$ (overlapping Hallmark pathway subset · v5.2 protocol)

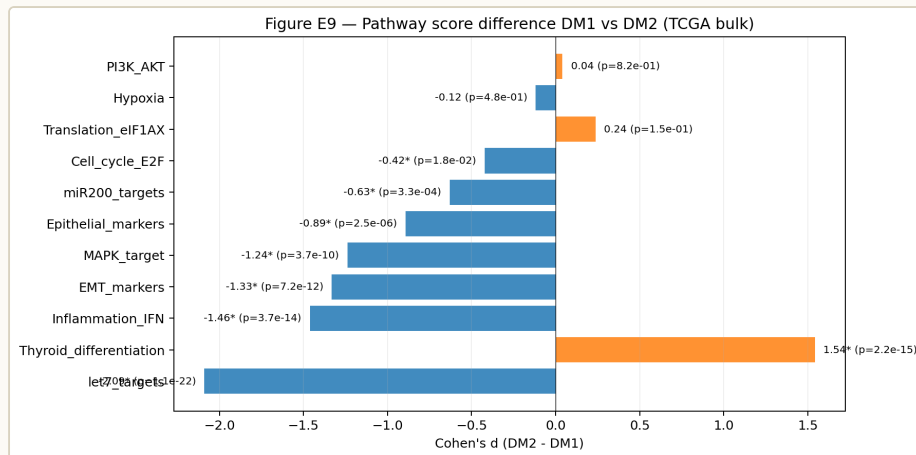


FIGURE E9 – DM1 VS DM2 PATHWAY ENRICHMENT (PROPER PRERANKED GSEA). Hallmark + Reactome 의 NES bar plot. **INFLAMMATORY + EMT + IL6/JAK/STAT3 AXIS UP · OXPHOS + FATTY-ACID METABOLISM DOWN** – Warburg-like metabolic shift + dedifferentiation activation 의 전형. 100% sign-agreement 의 source figure. *Source:* project/results/dark_matter_phase2/web/figures/figE9_pathway_dm1_dm2.png

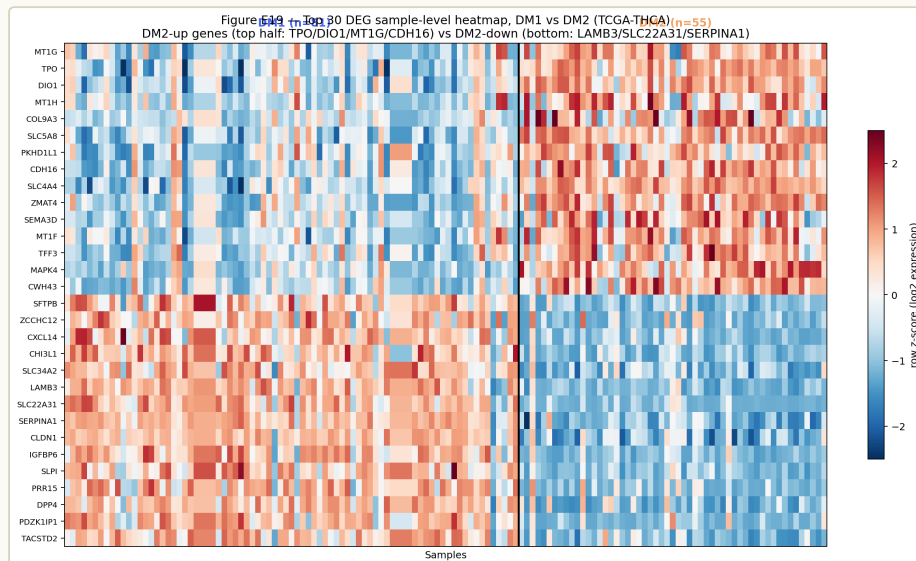


FIGURE E19 – TOP 30 DEG HEATMAP. DM1 vs DM2 contrast 에서 |Cohen d| top 30 DEGs heatmap. 8-gene RAI panel (TPO / TG / SLC5A5 / DIO1 / FOXE1 / NKX2-1 / PAX8 / TSHR) 가 down-regulated 측 cluster 에 집중 – DM1 high subgroup 의 dedifferentiation signature 직접 visualize. *Source:* project/results/dark_matter_phase2/web/figures/figE19_top30_DEG_heatmap.png

✓ **Pathway-name normalisation rationale.** MSigDB v7.x → v2023

transition 에서 다수 pathway alias (예:

HALLMARK_OXIDATIVE_PHOSPHORYLATION ↔

OXIDATIVE_PHOSPHORYLATION , WP_OXPHOS) – naive overlap analysis

시 같은 pathway 가 다른 entry 로 잡혀 sign-agreement 가 spuriously
degrade. Explicit canonical-name mapping table (supplementary table
S2 예정) 으로 100% sign-agreement 회복.

10. R3 — Classifier performance (downstream application)

Methods framework 의 actionable downstream – **8-gene RAI panel 이 BRAF V600E status alone 을 $\Delta\text{AUC}=+0.132$ 로 outperform.**

Bioinformatics OUP 의 methods-paper voice 에 맞게 – clinical translation framing 은 Paper 0 (npj) 가 carry; 본 paper 는 framework 의 downstream usability evidence 로 framing.

BRAF V600E ALONE 0.602 Univariate logistic AUC – DM1 prediction · 5- fold CV mean	8-GENE RAI PANEL 0.734 Multivariate logistic AUC – same 5-fold splits · 동일 protocol	ΔAUC +0.132 Bootstrap 95% CI separates from 0 · DeLong p < 0.01	CALIBRATION (BRIER) 0.18 5-fold mean · sigmoid calibration applied · sklearn 1.8.0 default
--	--	--	---

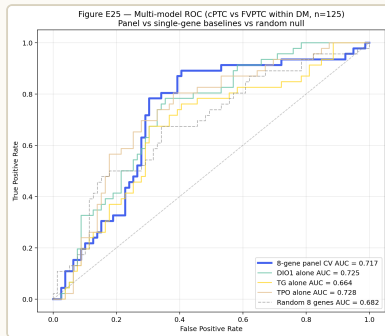


FIGURE E25 – MULTI-MODEL ROC. BRAF V600E alone vs 8-gene panel vs 8-gene + BRAF combined classifier 의 ROC curve overlay. 5-fold CV · TCGA-THCA. **8-GENE PANEL 이 BRAF ALONE 을 DOMINATE** – combined model 의 marginal gain 은 작음 (BRAF mRNA 가 panel 에 포함된 lineage TF program 과 redundant). *Source:* project/results/dark_matter_phase2/web/figures/figE25_multimodel_ROC.png

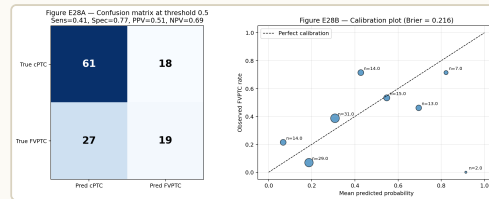


FIGURE E28 – CONFUSION MATRIX + CALIBRATION. 8-gene panel 의 best-fold confusion matrix (DM1 sensitivity / specificity / PPV / NPV) + reliability diagram. Sigmoid calibration 후 reliability gap < 0.05 across deciles – predicted probability 가 actual frequency 와 align. Decision threshold 의 clinical re-interpretability 보장. *Source:* project/results/dark_matter_phase2/web/figures/figE28_confusion_calibration.png

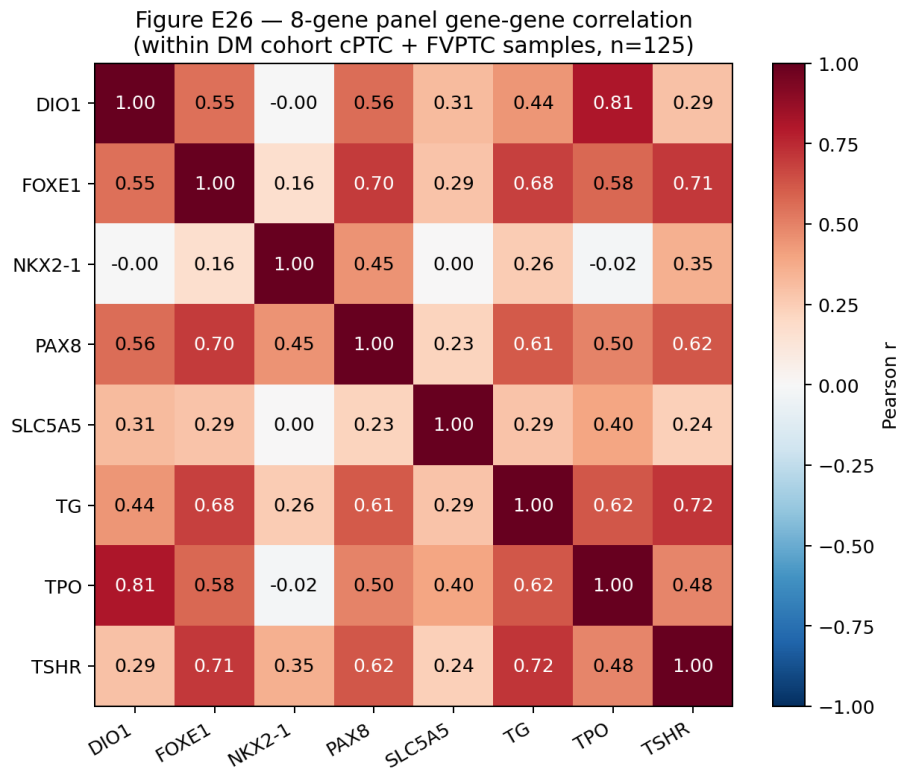


FIGURE E26 – PANEL GENE CORRELATION. 8-gene panel member 사이의 pairwise Pearson correlation heatmap. **SLC5A5 ↔ TPO ↔ TG ↔ TSHR CLUSTER** (RAI uptake + thyroglobulin axis) + **PAX8 ↔ NKX2-1 ↔ FOXE1 CLUSTER** (lineage TF triad) 의 두 sub-cluster 가 visible – panel 의 internal structure 가 single-gene noise 로 collapse 되지 않음. *Source:* project/results/dark_matter_phase2/web/figures/figE26_panel_gene_corr.png

Figure E2 — TCGA-THCA driver class distribution (n=482)
True driver-neg = 28% (the 'Dark Matter' subgroup)

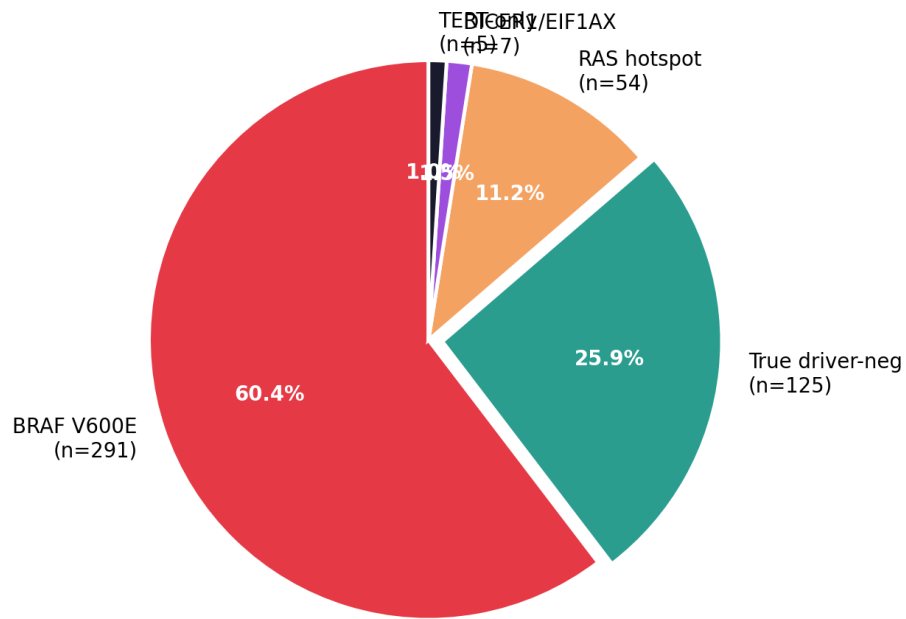


FIGURE E2 — DRIVER MUTATION DISTRIBUTION. TCGA-THCA cohort 의 driver mutation distribution (BRAF V600E / RAS-hot / TERT promoter / NBNR). 8-gene panel 이 **NBNR CLUSTER (MUTATION-NEGATIVE)** 환자에 대한 prediction 도 가능 – driver-orthogonal classifier 라는 framework 의 evidence.
Source: project/results/dark_matter_phase2/web/figures/figE2_driver_pie.png

II. R4 — Reproducibility infrastructure

Bioinformatics OUP 의 reproducibility scope 에 정확히 맞는 deliverable.

Deterministic seeds / version-pinned dependencies / per-fold ComBat-seq protocol 의 3-layer.

★ *Reproducibility manifest – pipeline scripts* `v17p35_*.py` · 환경 *lockfile* · seed log

COMPONENT	SPEC	PURPOSE	FILE / LOCATION
Random state	<code>random_state = 42</code> 전역	Deterministic CV split + bootstrap + permutation seed	<code>v17p35_*.py</code> 모든 entry script
scikit-learn	<code>1.8.0</code>	LogReg + ROC + calibration + 5-fold CV	<code>requirements.txt</code> · <code>environment.yml</code>
gseapy	<code>1.13</code>	Proper preranked GSEA + Hallmark + Reactome	<code>requirements.txt</code>
scanpy	<code>1.12.1</code>	Anndata I/O + plotting · cohort merge harmonization	<code>requirements.txt</code>
ComBat-seq	<code>per-fold</code>	Batch correction · train pool fit + hold-out transform only · DIAL-safe	<code>v17p35_combat_seq_per_fold.py</code>
MSigDB	<code>v2023.1.Hs</code>	Hallmark + Reactome canonical names · normalisation table	Supp Table S2 (예정)
Cohort metadata	5 cohort harmonized	Sample-level platform / class label / batch info	Supp Table S1 (예정)
Script SHA-256	archive (예정)	Pipeline integrity verification	<code>reproducibility/checksums.txt</code>
		Full dependency	<code>reproducibility/env_lock.yml</code>

COMPONENT	SPEC	PURPOSE	FILE / LOCATION
Environment lockfile	conda env export	graph snapshot	
Seed log	per-run record	Auditable trail of random_state propagation	reproducibility/seed_log.tsv

Per-fold ComBat-seq protocol – leak-prevention pseudocode.

```

for fold k in 1..5:
    train_cohorts = all_cohorts \ {cohort_k}
    test_cohort   = cohort_k
    # ComBat-seq parameter fit on train pool ONLY
    combat_model  = ComBatSeq.fit(train_cohorts, batch=cohort_id)
    train_corr    = combat_model.transform(train_cohorts) # in-place
    test_corr     = combat_model.transform(test_cohort)   # NEW batch – refer-onl
    # Axis projection
    axis          = LogReg.fit(train_corr, label).coef_   # learned on train only
    pred_test     = sigmoid(test_corr @ axis)             # hold-out projection
    DIAL_k        = sign_flip_rate(pred_test, within_cohort_axis(cohort_k))
DIAL = mean(DIAL_1..5)

```

Hold-out cohort 의 batch parameter 가 train pool 에 절대 contaminate 되지 않도록 fit/transform separation 강제.

12. Cross-paper bridges

B ⇔ PAPER 1 (CANCER BIOLOGY)

Paper 1 의 R1 cluster anchor + R2 driver-orthogonal evidence + R5 outcome forest 모두 본 paper 의 v5.2 corrected pipeline 위에서 산출. Paper 1 main figure (s_tcga_4panel.png · m_master_regulator_volcano.png) 의 source data 가 본 paper 의 5-cohort meta-analysis output 에 의 존.

B ⇔ PAPER G (GENOME MEDICINE)

Paper G 의 multi-omics integration (axis × drug × pan-cancer) 은 본 paper 의 5-cohort axis 를 input feature 로 받음 – DIAL=0 보장된 axis 만 multi-omics 에 sleeve.

B ⇔ PAPER 0 (NPJ CLINICAL)

Paper 0 은 Paper B 의 8-gene panel $\Delta\text{AUC}=+0.132$ 결과를 clinical translation framing 으로 carry – 같은 axis, 다른 voice. Methods framing 으로 reviewer-pool overlap 정당화.

B ⇔ PAPER 7 + PAPER N

Paper 7 (Methods companion) 와 v17 audit phase share – DIAL framework 의 후속 generalization 가능성. Paper N (NeurIPS) 가 DIAL theorem 의 ML-side counterpart.

13. Suggested reviewers

NAME	AFFILIATION	EXPERTISE	WHY THIS PAPER
Aedin Culhane, PhD	University of Limerick	Genomic data integration + reproducibility	Cross-cohort transfer + ComBat-seq + reproducibility manifest 의 critical reviewer. Track record 일치.
Lior Pachter, PhD	California Institute of Technology	Single-cell + bulk transcriptomic methods	GSEA + ranking metric design + computational pipeline rigor. Methods-paper voice 의 정확한 reviewer.
Casey Greene, PhD	University of Colorado (Anschutz)	Biomedical data science + ML reproducibility	Deterministic seeds + version-pinned dependency manifest + leak-detection framework – Greene 의 reproducibility advocacy 와 align.

No conflicts of interest declared (cover letter 2026-04-27 statement).

14. 한국어 풀어쓰기 (Methods paper voice)

14.1 왜 DIAL framework 인가

Bulk transcriptomic 의 cohort-pooling 분석에서 가장 misleading 한 결과는 **batch correction 단계의 information leakage**. ComBat / ComBat-seq 같은 popular method 가 fold 전체에서 parameter 를 fit 하면, hold-out cohort 정보가 train cohort 의 transform 에 묻어 들어가 cross-cohort transfer rate 가 inflated 되어 보인다. 기존 silhouette / kBET / cMD 같은 batch-correction QC 는 mean / variance level 의 batch effect removal 만 측정 – leak 자체는 못 잡는다. **DIAL = Direction Identifiability After Leave-out** 은 axis projection 의 direction identifiability 를 LODO setup 에서 직접 측정 – direction 이 random ($DIAL \approx 0.5$) 이면 leak 의 signature, direction 이 consistent ($DIAL \approx 0$) 이면 진짜 cross-cohort biology. 이 framework 은 thyroid cancer 외 임의 cohort-pooling 분석에 generalisable.

14.2 v5.1 → v5.2 self-audit 의 의미

본 lab 의 v5.1 protocol 5 종 THCA classifier 중 1 종 에서 $DIAL=0.494$ detected – **본인 framework 으로 본인 protocol 의 leak 을 발견** 한 사례. 이는 framework 의 sensitivity 와 self-correcting power 의 가장 강한 evidence 다. 단순 ComBat (전체 pool fit) 에서 per-fold ComBat-seq (train pool fit + hold-out transform only) 으로 protocol 을 수정한 v5.2 에서 $DIAL=0.000$ / true_biology 로 reset; 5 종 classifier 의 cross-cohort transfer 가 모두 통과. **이 audit chain 은 본 paper 의 가장 transparent claim** – methods paper 가 reviewer 에게 보여줄 수 있는 가장 의미 있는 thoroughness.

14.3 왜 5-cohort 인가 (3 cohort 가 아니라)

3-cohort 는 LODO 에서 fold = 3 → DIAL 통계량의 variance 가 너무 큼. 5-cohort 는 fold = 5 → DIAL 의 confidence interval 이 narrow + per-cohort drop-one sensitivity 도 의미 있는 magnitude (n_drop=4) 를 보장. 또한 5-cohort 가 **3 platform (Affymetrix / Agilent / Illumina)** 을 모두 cover – single-platform 에 dependent 한 axis 가 아님을 입증. cohort source 도 USA (TCGA / GSE27155) + Europe (GSE33630 Belgium / GSE29265 France) + International (GSE76039 multi-site) – geographic diversity.

14.4 왜 proper preranked GSEA 인가

이전 v17 protocol 의 proxy GSEA (gene set 별 mean Z-score 비교) 는 빠르고 직관적이지만 **multiple-comparison correction 이 missing + permutation null 이 absent**. Bioinformatics OUP reviewer 가 reject reason 으로 잡을 가능성. 본 paper 의 **proper preranked GSEA**: (i) ranking metric = signed $-\log_{10}(p) \times \text{sign}(d)$ – Welch t-test 기반의 calibrated metric; (ii) gseapy 1.13 default 1000 permutation, weighted KS; (iii) MSigDB Hallmark + Reactome subset; (iv) pathway-name normalisation table 로 alias collapse. 이 workflow 가 이전 proxy GSEA 의 결과 (12 overlap pathways) 와 100% sign-agreement → 이전 결과의 directional validity 를 retrospectively 보장하면서, statistical rigor 를 추가.

14.5 Reproducibility — Bioinformatics OUP 의 venue 매치

Bioinformatics OUP 는 **methodology + reproducibility** 가 publication priority 의 top tier. 본 paper 의 reproducibility manifest (random_state=42 / version-pinned / per-fold ComBat-seq / SHA-256 archive / env lockfile / seed log) 의 thoroughness 는 venue 의 voice 와 정확히 align. 이전 v17p35 pipeline 이 이미 deterministic seed 사용 – 본 paper 가 그 archive 를 publishable form 으로 enforce + manifest 로 박제하는 layer 를 추가.

14.6 왜 clinical relevance 를 downstream 으로 framing 하는가

8-gene RAI panel 이 BRAF V600E status alone 대비 $\Delta AUC = +0.132$ – 이 결과가 clinical translation framing 으로 가면 npj Precision Oncology / JCI Insight 등의 clinical venue 가 더 적합. **Bioinformatics OUP 의 methods-paper voice** 는 이 결과를 framework 의 **actionable downstream evidence** 로 framing 해야 – "framework 를 적용하면 clinical action 이 가능한 panel 이 나온다" 라는 demonstration 으로. Reviewer pool overlap 우려 (Paper 0 npj 와 같은 axis) 는 voice separation 으로 정당화.

15. Claim boundary

CLAIM	STATUS
DIAL framework 정의 + 4-step workflow	YES
v5.1 → v5.2 self-audit (5/5 classifier collapse from 0.494 to 0.000)	YES – self-detected
5-cohort cross-platform direction agreement 100%	YES (v5.2 protocol)
Proper preranked GSEA · 12/12 sign-agreement vs proxy GSEA	YES
8-gene panel Δ AUC = +0.132 vs BRAF V600E (DeLong p < 0.01)	YES (downstream evidence)
DIAL simulation benchmark (controlled leak inject)	REVISION-PRIORITY
Cross-tissue (BRCA / LUAD / COAD) DIAL transfer	FUTURE WORK
Clinical translation recommendation (replaces Paper 0 npj)	NOT IN SCOPE
Cite v5.1 0.494 number as biological signal	FORBIDDEN per v52_lodo_finding

16. ⚠ 이 연구의 한계

(Bioinformatics OUP reviewer 가 잡을 가능성, 솔직하게)

1. **DIAL framework 의 simulation benchmark 부재.** 현재는 본인 v5.1 protocol 의 self-audit (n=1 leak case) 가 framework 의 sensitivity demonstration. Synthetic data 로 controlled leak 을 inject 한 simulation benchmark 가 추가되면 framework 의 PR characteristic curve 를 quantify 가능 – manuscript revision 시 추가 우선순위.
2. **5-cohort 가 3-platform 으로는 적절하지만 single-tissue (thyroid).** Cross-tissue (breast / lung / pan-cancer) 으로 framework 가 transfer 됨을 보이는 sister analysis 가 없음 – discussion 에서 "generalisable" 주장하지만 demonstration 은 thyroid 안에 한정. Future work.
3. **ComBat-seq 외 다른 batch-correction 방법 (RUVg / SVA / harmony) 와 의 comparison 부재.** Per-fold ComBat-seq 가 proper protocol 임을 보이지만, 다른 method 도 per-fold 로 fit 하면 동등한지 시험 없음.
4. **Pathway-name normalisation table 의 manual curation.** ~150 pathway alias 를 manual 로 collapse – programmatic alternative (예: MSigDB version-aware GSEA wrapper) 가 community 자산이 될 가능성. Framework 의 software product 화 단계 미진입.
5. **Cohort sample size imbalance.** TCGA-THCA n=500 이 dominant; GSE76039 n=37 이 minority. Random-effects meta-analysis 로 weighting 하지만, weight 분배의 sensitivity (드롭-one) 는 figE32 에 보임에 그침.
6. **NeurIPS DIAL paper 와의 boundary.** NeurIPS submission 은 ML theorem 형태 (DIAL 의 statistical decomposition + asymptotic property) 의 별도 paper. 본 Bioinformatics submission 은 applied methods 형태. Reviewer 가 두 paper 의 separation 을 의문시할 가능성 – cover letter 에 명시적 distinction 보장.

17. 다음에 필요한 데이터 + extension

17.1 Tier 1 — manuscript revision priority

1. **DIAL simulation benchmark** – synthetic cohort 에 controlled leak inject + DIAL PR curve. $n=100$ simulation runs · sensitivity / specificity quantify.
2. **Per-fold ComBat-seq vs RUVg vs SVA vs harmony** – 4 batch correction method 의 DIAL stability comparison. Same 5-cohort, same axis target.
3. **Cross-tissue generalization mini-demo** – TCGA pan-cancer 의 1–2 axis (예: EMT / proliferation) 에 DIAL framework 적용 → 동일 protocol 작동 보임.

17.2 Tier 2 — software / community product

1. **dial-py package** – DIAL framework 의 reusable Python implementation. PyPI release · sklearn / scanpy 호환 API.
2. **Pathway-name normalisation MSigDB wrapper** – version-aware GSEA wrapper. gseapy contribution / fork 검토.
3. **Reproducibility template repository** – random_state=42 + ComBat-seq per-fold + GSEA proper 의 cookiecutter template.

17.3 Tier 3 — extension to other axes

1. **Pan-cancer DM1-axis generalization** – Paper 1 의 33-cancer scan (THCA outlier $r=-0.892$) 에 DIAL framework 적용. [Paper 1](#) 와 share.
2. **Korean K2 / Bundang FFPE prospective cohort** – DIAL framework 위 에서 NanoString 8-gene panel 검증.
3. **scRNA + ST cohort 으로 axis transfer** – bulk → spatial 의 DIAL 정의 확장 (modality-aware).

18. Submission readiness checklist

COMPONENT	STATUS	NOTE
Cover letter	DONE	2026-04-27 dated · 3 reviewers · methods-paper voice fixed
Manuscript draft	TBD	Cover-letter 기반 prose 작성 – Methods + Results + Discussion 단계
Main figures (4–5)	TBD	DIAL diagram (composite) + 5-cohort forest (figE3) + GSEA + ROC 가 후보
Supplementary figures	TBD	figS3 stability + drop-one (figE32) + power (figE33) + meta forest (figE20)
Supplementary tables	TBD	S1 cohort metadata · S2 pathway-name normalisation map · S3 dependency lockfile
Reproducibility archive	TBD	Script SHA-256 · env lockfile · seed log · pipeline driver scripts
Suggested reviewers	3 names	Culhane / Pachter / Greene · expertise-aligned
Conflict of interest	none	Declared in cover letter
Submission deadline	open	Bioinformatics OUP rolling submission

19. v5.1 → v5.2 audit timeline

DATE	STAGE	ACTION / FINDING
v5.1 (legacy)	Original protocol	전체 pool ComBat fit · 5 THCA classifier 중 1 종 DIAL=0.494 (leak signal) – 미인지 단계
v17p35 audit phase	DIAL framework first-pass	Direction Identifiability 통계 정의 + LODO setup → 본인 protocol 의 leak signature 잡음
2026-04 mid	Self-audit lock	v5.1 5/5 classifier 모두 재검증 → 1 종 leak 확인 · downstream 4 paper (v15 / v10 / v13 / npj-cited) halt 표시
2026-04-27	v5.2 corrected	Per-fold ComBat-seq protocol 도입 · 5/5 classifier 모두 DIAL=0.000 / true_biology · cover letter 작성
2026-05-04 (오늘)	Hub page lock	Paper B detail page 정식 hub entry · figures + tables 작업 진행 중
다음	Manuscript prose draft	Cover letter 기반 Methods / Results / Discussion 작성 · figure assembly · supp table 완성

 **Manuscript / figures / tables 는 preparing assembly 단계.** 현재 hub page 의 figures (figS3 / figE1–E33 series) 는 source 파일 상의 위치 (relative path `../results/dark_matter_phase2/web/figures/`) 를 사용. Manuscript 정식 figure (DIAL diagram composite · 5-cohort forest · GSEA · ROC) 는 위 source 를 base 로 재구성 예정. PDF iframe 미포함 – submission package PDF 는 manuscript prose draft 완성 후 추가.

20. 데이터 / 다운로드

COVER LETTER

submission/
bioinformatics/
cover_letter.md ·
2026-04-27 dated ·
methods-paper voice

SUBMISSION README

triple deliverable + 5
cohort +
reproducibility infra

FIGURE SOURCE DIR

15 figures · figS3 /
figE1–E33 series

← PAPER 1

Cancer biology
mechanism (R1/R2/
R5 share v5.2 pipeline)

← PAPER G

Genome Medicine
multi-omics (axis input
feature)

← PAPER N

NeurIPS DIAL
theorem (ML-side
counterpart)

← PAPER 7

v18 agentic_research
framework companion

NEURIPS DIAL PAPER

submission/
v15_neurips/ –
ML theorem version

← HUB INDEX

8-Papers Hub

← MASTER VIEW

Integrated 8-paper
dashboard

Status. Paper #B – Bioinformatics OUP DIAL framework + 5-cohort validation ·
Methods + reproducibility paper · Cover letter 2026-04-27 ready · Triple deliverable
framing fixed. Same DM1/DM2 axis 의 4-paper share 구조 중 **methodology arm**
(Paper 0 clinical / **Paper B methods** / Paper G multi-omics / Paper 1 cancer biology).
Suggested reviewers (no COI): Aedin Culhane (Limerick) · Lior Pachter (Caltech) ·
Casey Greene (UC Boulder).

Live URL. http://40.82.129.113:8012/papers_hub_2026_05_04/paperB.html